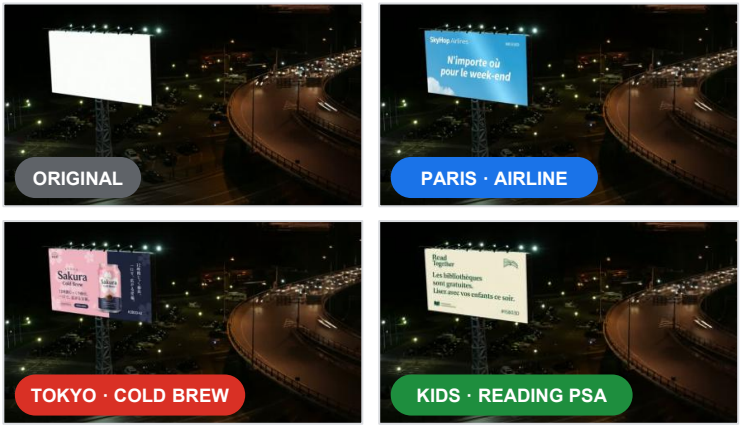
Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Team Details

a. Team name: Berkians
b. Team leader name: Sparsh Srivastava (Team member: Rajendra Sarpal)
c. Problem Statement: Ad breaks interrupt streaming and get skipped. InScene places brand-safe, personalised ads inside the scene itself, rendered by Gemini for every viewer.



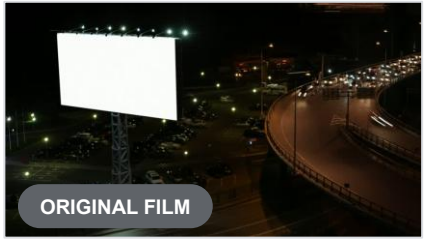
Brief about the idea

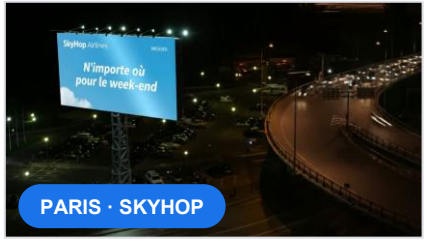
InScene: the ad is in the scene.

Streaming audiences skip, mute and resent ad breaks, yet brands keep paying for them. InScene removes the break: it finds real surfaces inside a film (billboards, screens, signs) and uses Gemini to repaint them with a brand-safe, localised ad chosen for each viewer.

The same scene plays differently for Camille in Paris (an airline, in French), Aiko in Tokyo (cold brew, in Japanese) and Srikant in Bengaluru (Diwali coffee, in Kannada). Kids see a library PSA; ad-free subscribers see clean art.

Directors stay in control of every slot, brands brief once for every market, and viewers can pause any scene to shop it with a QR code.









One scene. Every viewer. A different ad, and the story never stops.

Gemini finds the surface, picks a brand-safe ad per viewer and repaints it into the film.

Opportunities



1 How is it different?

- Classic product placement is fixed at shoot time: one brand, forever, for everyone.
- Virtual-ad tools need VFX teams per market. InScene is self-serve and per viewer.
- Server-side ad insertion still breaks the story; InScene never stops the film.

2 How does it solve the problem?

- Gemini scouts ad surfaces and reads each scene's mood for brand safety.
- Gemini Omni repaints only the slot, so the rest of the frame is untouched.
- Each cut is measured for visibility and checked against the director's cap before it plays.

3 USP

- One film, a different ad per viewer: language, city, season, age, plan.
- Directors approve every slot; nothing plays without sign-off.
- Pause to shop: every in-scene ad becomes a QR checkout.
- Seasonal: Diwali or Cherry Blossom campaigns switch on automatically.

List of features offered by the solution

Slot Scout

Gemini finds billboards, screens and signs in any uploaded film and scores each one.

Brand safety

Gemini reads mood, themes and sensitive context; brands list scenes to avoid.

Director control

Blocked categories, prominence cap and approval before any cut plays.

Brand Studio

One brief: logo, product, markets, languages, season and targeting.

Per-viewer cuts

Gemini Omni repaints the slot for each viewer's city, language and age.

Kids & ad-free

Kids get PSAs only; ad-free plans get neutral artwork, never an ad.

Audio ads

Localised voice ads (Gemini TTS) mixed into the scene's soundtrack.

Pause to shop

Pausing shows what's in the scene with a QR code to buy.

Why this ad?

Every placement explains why it won and which campaigns did not.

Exposure score

Gemini measures frame share, visibility and legibility before playback.

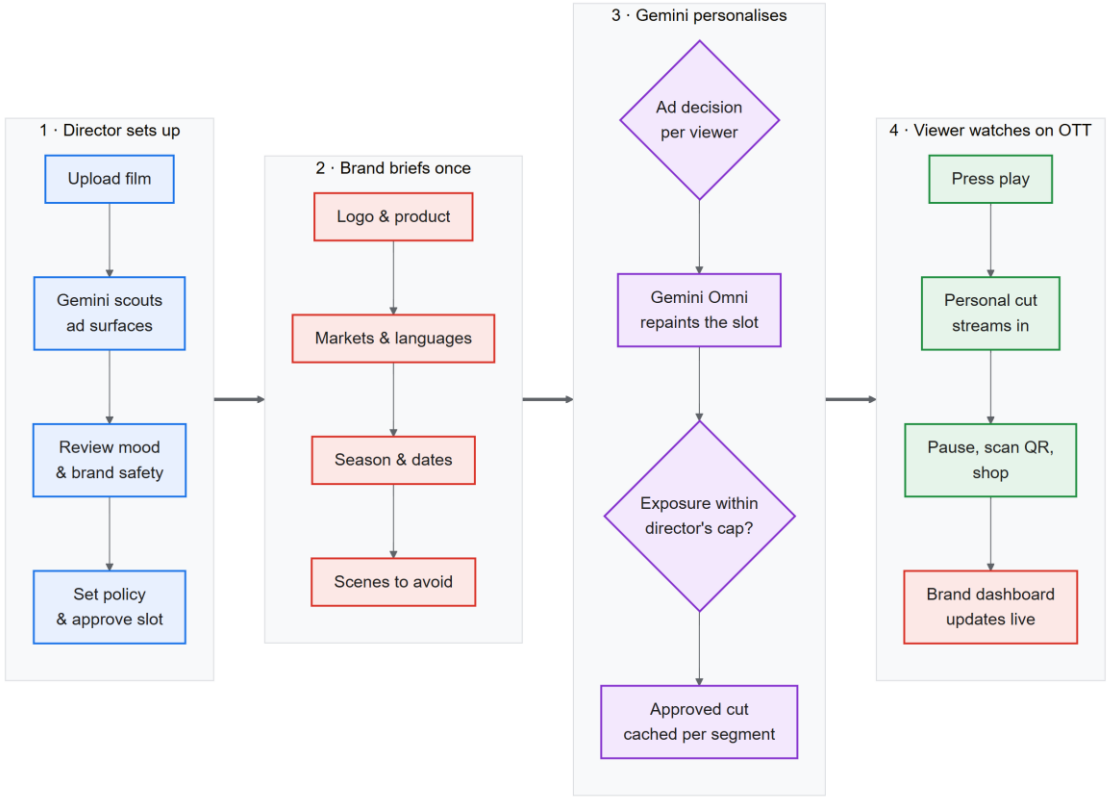
Seasonal calendar

Campaigns follow dates and festivals, e.g. Diwali in India.

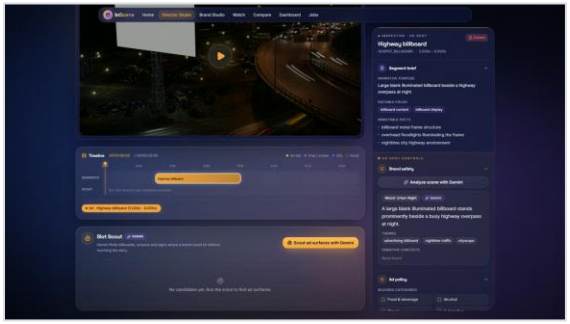
Brand dashboard

Live impressions, pause views, clicks and audio plays per city.

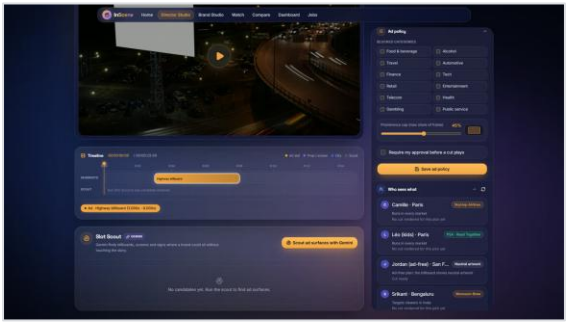
Process flow diagram or Use-case diagram



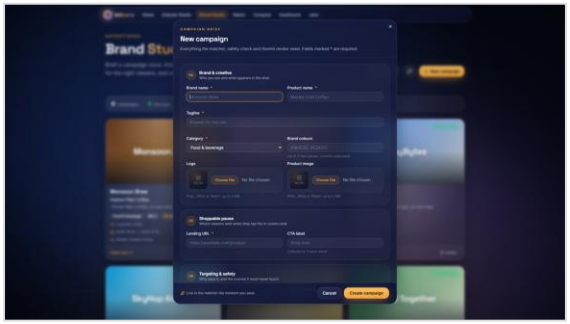
Wireframes/Mock diagrams of the proposed solution (optional)



Director Studio: Slot Scout



Director policy & approval

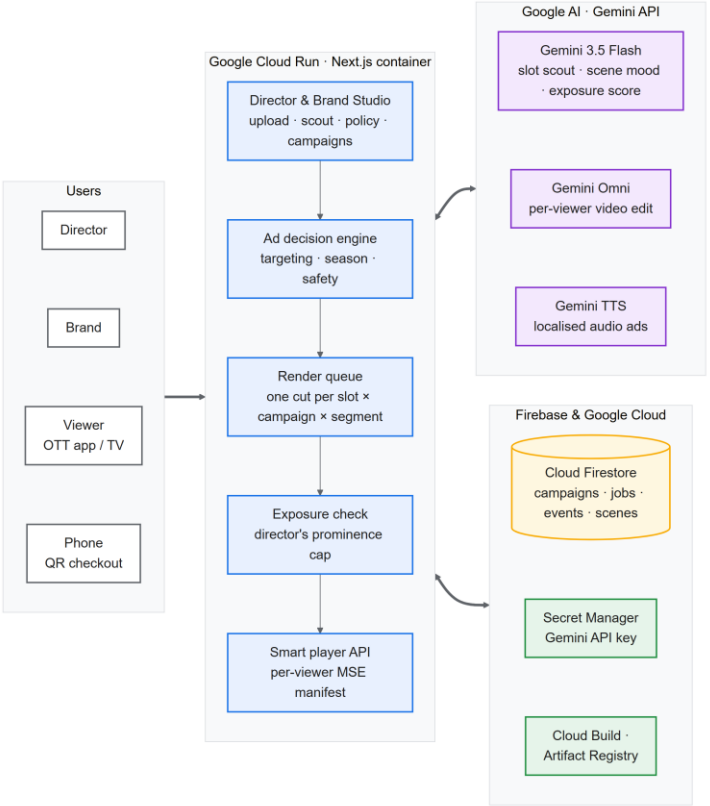


Brand Studio: one brief



Viewer profiles

Architecture diagram of the proposed solution



How a cut is made

1. Director uploads a film; Gemini 3.5 Flash scouts ad surfaces and reads scene mood.

2. Brands brief once; campaigns live in Cloud Firestore.

3. The ad engine picks an ad per viewer segment; Gemini Omni repaints the slot.

4. Gemini scores exposure against the director's cap before anything plays.

5. The player stitches the approved cut into the film live; events flow back to Firestore for the brand dashboard.

Everything runs in one Cloud Run container; the Gemini key comes from Secret Manager.

Technologies to be used in the solution

Google AI

- Gemini 3.5 Flash: ad-surface scouting, scene mood & safety, exposure scoring
- Gemini Omni (video edit): repaints the ad slot per viewer
- Gemini 2.5 Flash TTS: localised audio ads
- Google Gen AI SDK (@google/genai)

Google Cloud & Firebase

- Cloud Run: serverless container for the whole app
- Cloud Firestore via Firebase Admin SDK: campaigns, render jobs, analytics events, scene analyses
- Cloud Build + Artifact Registry: container builds
- Secret Manager: API keys

App & media

- Next.js 16 (App Router), React 19, TypeScript
- Tailwind CSS 4, Framer Motion, React Three Fiber (3D hero)
- Media Source Extensions player that stitches per-viewer segments
- FFmpeg for segment export; Vitest for tests

Estimated implementation cost (optional)

Cost scales with campaigns, not with viewers

Each render is keyed by slot × campaign × audience segment, cached, and reused for every matching viewer.

4 ad slots in a film × **10** audience segments (city × language) = **40** short Gemini Omni renders, total

…the same 40 renders whether 1,000 or 1,000,000 people watch.

Scales to zero

Cloud Run idles at no cost between sessions; Firestore stays inside the free tier at prototype traffic.

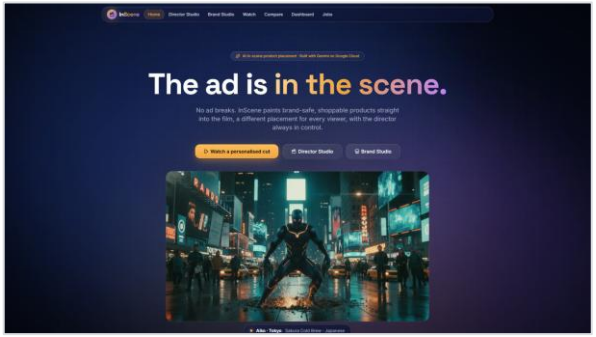
Analyse once

Gemini Flash scene analysis runs once per slot and is cached, never per viewer.

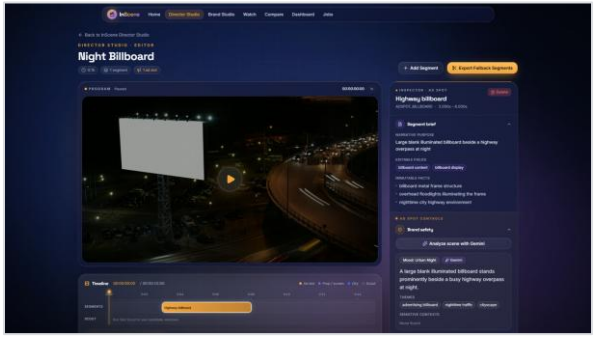
Never a broken stream

An unrendered or rejected cut falls back to the original scene, so a failed render costs nothing on screen.

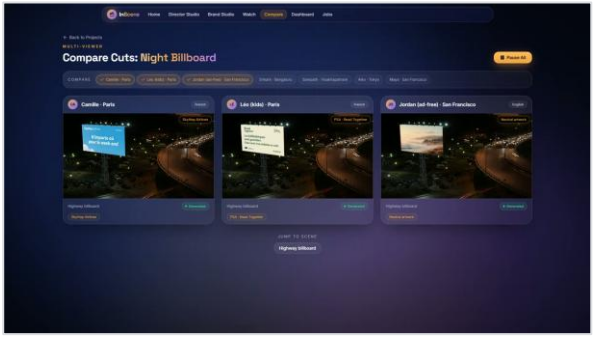
Snapshots of the prototype



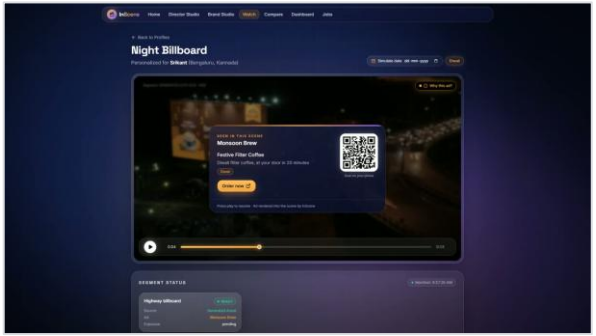
Home



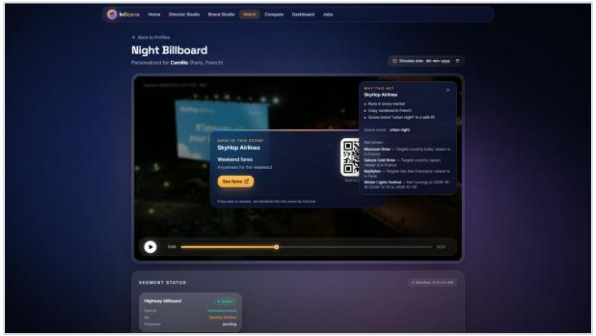
Director Studio



Compare: one scene, three viewers



Pause to shop (Bengaluru, Diwali)



Why this ad?



Brand dashboard

Prototype Performance report/Benchmarking

~1.5 min
Gemini Omni render per 5 s ad slot (measured)

0
reloads: per-viewer segments stitched live with MSE

73 / 100
exposure score, Paris cut: 9% of frame, under the 45% cap

10
real per-viewer renders across 2 films and 7 profiles

1×
scene analysis per slot, cached for every viewer

31 / 31
unit tests passing (matching, safety, seasons)

Measured on the prototype with real footage. A viewer's first play uses a ready cut; while a cut is rendering, or if it was rejected by the exposure check, the original scene plays instead, so playback never breaks.

Additional Details/Future Development (if any)

1 Live sports & streams

Repaint stadium boards and LED screens in near real time.

2 Studio integrations

Plug-ins for OTT players and ad servers (VAST/VMAP).

3 Attention analytics

On-device attention signals, privacy first.

4 Creative generation

Brands upload a logo; Gemini designs the in-scene art per market.

5 More surfaces

Bus sides, packaging, phone screens and product swaps.

6 Revenue share

Automatic payouts to studios per placement and view.

Provide links to your:

- GitHub Public Repository: https://github.com/Rajendra465/aibuildercup-sub
- Demo Video Link (3 Minutes): https://youtu.be/Os0ZW7ywv_M
- Final Product Link: https://inscene-755276882523.us-central1.run.app

Watch the demo

youtu.be/Os0ZW7ywv_M

Try the live app

inscene-755276882523.us-central1.run.app

Read the code

github.com/Rajendra465/aibuildercup-sub

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

